

AIDATACENTER — BOOTSTRAP STACK

AI Cluster Bootstrap Stack

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BOOT-rack platform for the 64× B200 inference cluster ([SPEC.md](#) §2A, §7).

1 Layers

Phase	What	Where
−1 — SoT	NetBox DCIM/IPAM/cabling (seed Docker → later K8s)	netbox/
0 — Seed	DHCP, DNS, NTP, iPXE (leases from NetBox export)	seed/
1 — Bare metal	Metal3 + Cluster API → 3 CP + 8 GPU workers	capi/
2 — GitOps root	Flux installs platform from this repo	platform/
3 — Platform	NetBox HA, Registry, Vault, obs, Redpanda, GPU Operator	platform/apps/

B00T rack

seed01 (dnsmasq, matchbox, ironic) spines 7060DX5-64S x2	cp01..cp03 (etcd + API)	hub / utility (Harbor, Vault, Flux source)
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| mgmt + fabric



GPU-1 / GPU-2 workers (worker01..08 × 8 B200)

2 Quick start (order of operations)

```
# NetBox SoT (seed Docker) — or offline from seed/site.yaml
cd netbox && cp env.example .env && docker compose up -d
export NETBOX_URL=http://127.0.0.1:8081
export NETBOX_TOKEN=$(bash scripts/create_token.sh)
pip install -r requirements.txt && python3 scripts/import_seed.py
bash ../scripts/netbox-sync.sh
# offline alternative: bash scripts/netbox-sync.sh --offline

# On seed host (Ubuntu 24.04 LTS, B00T rack)
sudo bash scripts/00-seed-host.sh

# From operator workstation with kubeconfig to seed/management cluster
bash scripts/01-install-capi-metal3.sh
BMC_USERNAME=... BMC_PASSWORD=... bash scripts/02-apply-cluster.sh
bash scripts/03-install-flux.sh # after CP is Ready
```

Full procedure: [docs/bootstrap.md](#) · [NetBox: docs/netbox.md](#) · [Cabling: docs/cabling.md](#).

3 Inventory (generated)

Do not edit `inventory/cluster.yaml` — export from NetBox:

```
bash scripts/netbox-sync.sh          # live API
bash scripts/netbox-sync.sh --offline # from netbox/seed/site.yaml
```

4 Design choices

Decision	Choice	Rationale
Source of truth	NetBox DCIM/IPAM/cables	Single ops DB for inventory, IPAM, cabling guide
Provisioning	Metal3 + Cluster API	Matches SPEC Metal3/MAAS path; pure K8s, no MAAS ISAM lock-in
OS (CP + utility)	Ubuntu 24.04 (Ironic image)	Drivers, GPU Operator, enterprise familiarity
OS (GPU)	Ubuntu 24.04 + GPU Operator host driver	NVIDIA DCGM / cuda-compatible path
GitOps	Flux v2	Pull-based, multi-tenancy later
Secrets	Vault + External Secrets	BOOT rack duty per SPEC
Registry	Harbor	Air-gapped / mirror for GPU images
Observability	kube-prometheus-stack + Loki + Alloy	SPEC §11
Bus	Redpanda (3 replicas on BOOT/util)	Inference archive buffer SPEC §6.7

5 Directory map

```
bootstrap/  
  netbox/          # SoT: docker day-0, seed/site.yaml, import/export scripts  
  inventory/       # GENERATED from NetBox (cluster.yaml, ipam.yaml)  
  seed/            # pre-K8s host config (dnsmasq, matchbox, chrony)  
  capi/            # Cluster API Cluster, KubeadmControlPlane, Metal3MachineTemplate  
  platform/        # Flux kustomizations + HelmReleases (includes netbox HA)  
  scripts/         # ordered operator scripts + netbox-sync.sh
```